

24/09/26

Signal Processing

Announcements:

- Recap: DTFT properties, example: $x[n] = a^n u[n]$, $|a| < 1$
- DTFT theorems: Linearity, delay*, modulation*, time-reversal, convolution*, differentiation*
- Periodicity
- DTFT properties:

$$x^*[n] \longleftrightarrow X^*(e^{-j\omega})$$

$$x^*[-n] \longleftrightarrow X^*(e^{j\omega})$$

$$\text{Re}\{x[n]\} \longleftrightarrow X_e(e^{j\omega}) = \frac{1}{2} (X(e^{j\omega}) + X^*(e^{-j\omega}))$$

$$j \text{Im}\{x[n]\} \longleftrightarrow X_o(e^{j\omega}) = \frac{1}{2} (X(e^{j\omega}) - X^*(e^{-j\omega}))$$

$$x_e[n] = \frac{1}{2} (x[n] + x^*[n]) \longleftrightarrow X_R(e^{j\omega}) = \text{Re}\{X(e^{j\omega})\}$$

$$x_o[n] = \frac{1}{2} (x[n] - x^*[n]) \longleftrightarrow X_o(e^{j\omega}) = j \text{Im}\{X(e^{j\omega})\}$$

Example!

$$\text{Real } x[n] : x[n] = a^n u[n] ; a \text{ is real, } |a| < 1$$

$$\text{DTFT}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n \cdot u[n] \cdot e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n \cdot e^{-j\omega n} = \sum_{n=0}^{\infty} (a e^{-j\omega})^n$$

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}} = \frac{1}{1 - a \cos \omega + j a \sin \omega} = \frac{1 - a \cos \omega - j a \sin \omega}{(1 - a \cos \omega)^2 + a^2 \sin^2 \omega} \quad (1)$$

- Check for conjugate symmetry: $X(e^{j\omega}) = X^*(e^{-j\omega})$

$$X^*(e^{-j\omega}) = \frac{1 - a \cos(-\omega) + j a \sin(-\omega)}{1 - 2a \cos(\omega) + a^2 \cos^2(\omega) + a^2 \sin^2(\omega)} = \frac{1 - a \cos \omega - j a \sin \omega}{1 - 2a \cos \omega + a^2 - 2a \cos \omega + a^2 \cos^2 \omega + a^2 \sin^2 \omega} \quad (2)$$

(1) = (2) $\Rightarrow$ DTFT of $x[n]$ is conjugate symmetric.

- Check for evenness of $|X(e^{j\omega})|$

$$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})}$$

$$= \frac{1}{(1+a^2-2a\cos\omega)} \sqrt{(1-a\cos\omega)^2 + a^2\sin^2\omega} \quad - (3)$$

$$|X(e^{-j\omega})| = \frac{1}{1+a^2-2a\cos(-\omega)} \sqrt{(1-a\cos(-\omega))^2 + a^2 \underbrace{\sin^2(-\omega)}_{(-\sin\omega)^2}} \quad - (4)$$

$$(3) = (4).$$

- DTFT Theorems:

* Note: we will discuss the DTFT definition in more detail in the coming week.

$$\text{DTFT } \{x[n]\} = \sum_{n=-\infty}^{\infty} x[n] e^{j\omega n}$$

- Linearity: $a x[n] + b y[n] \longleftrightarrow a X(e^{j\omega}) + b Y(e^{j\omega})$

- Delay: $x[n-n_d] \longleftrightarrow e^{-j\omega n_d} X(e^{j\omega})$

- Modulation: $e^{j\omega_0 n} x[n] \longleftrightarrow X(e^{j(\omega-\omega_0)}) \checkmark$

- Time-reversal: $x[-n] \longleftrightarrow X(e^{-j\omega})$

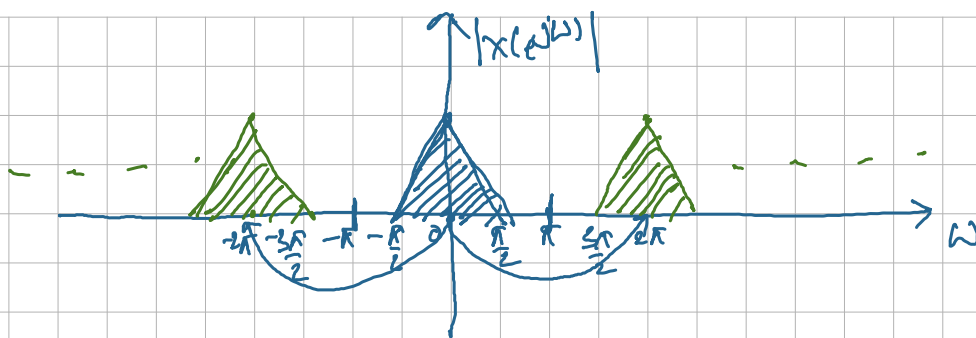
- Periodicity: $X(e^{j\omega}) = \sum x[n] e^{-j\omega n}$

if $\omega \rightarrow \omega + 2\pi$

$$X(e^{j(\omega+2\pi)}) = \sum x[n] e^{-j(\omega+2\pi)n}$$

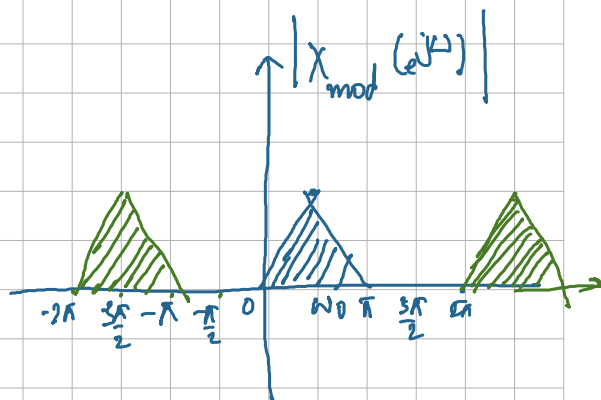
$$= \sum x[n] e^{-j\omega n} \underbrace{e^{-j2\pi n}}_1$$

$$= X(e^{j\omega}) \quad - (5)$$



- $x_{\text{mod}}[n] = e^{j\omega_0 n} x[n]$. Draw the magnitude spectrum of $x_{\text{mod}}[n]$.

$$\begin{aligned}
 X_{\text{mod}}(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_{\text{mod}}[n] \cdot e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{\infty} x[n] \cdot e^{j\omega_0 n} \cdot e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{\infty} x[n] \cdot e^{-j(\omega - \omega_0)n} \\
 &= X(e^{j(\omega - \omega_0)})
 \end{aligned}$$



- This is important in several applications including communications.